Manipal School of Information Sciences

Linux Operating System Lab

LAB ASSIGNMENT 02

Linux Laboratory Assignment 02

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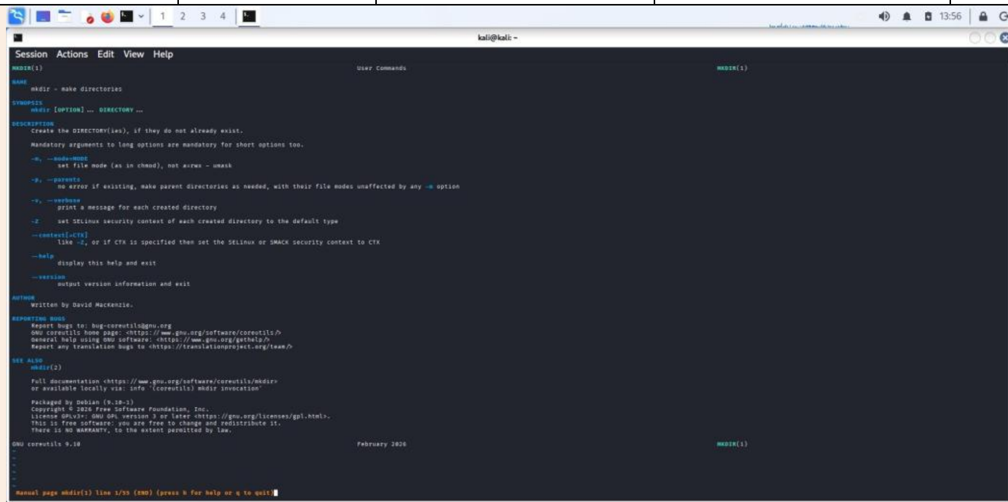
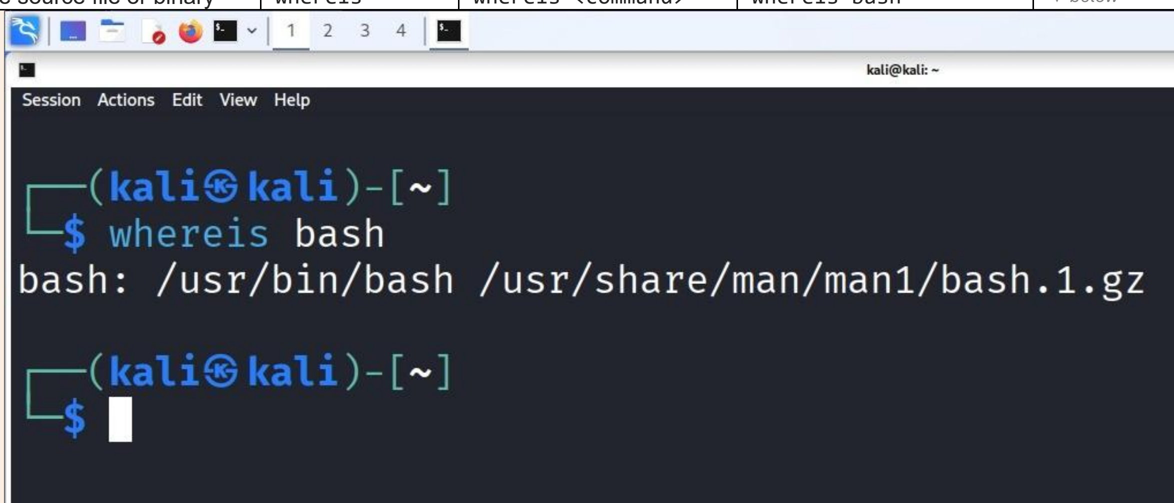
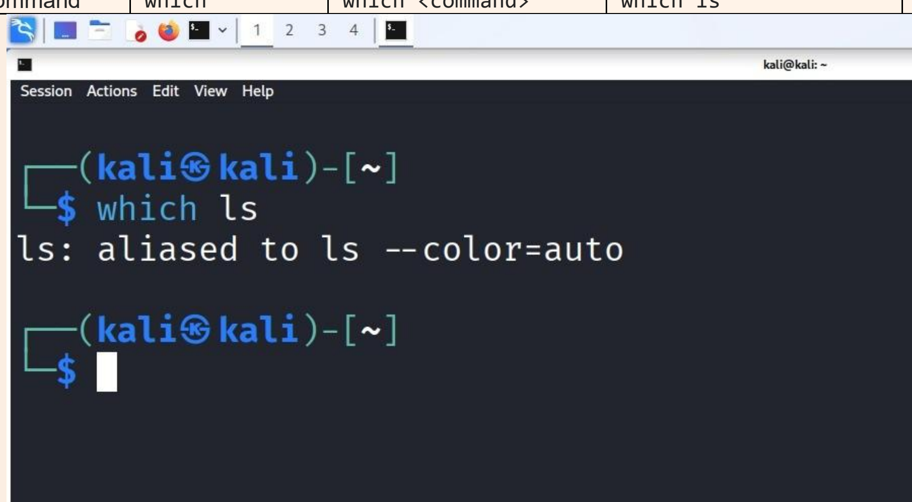
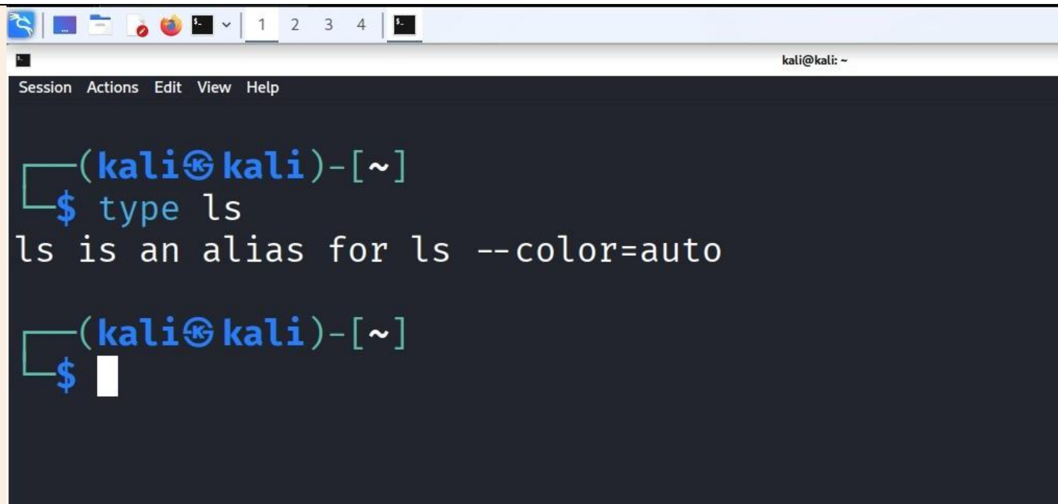
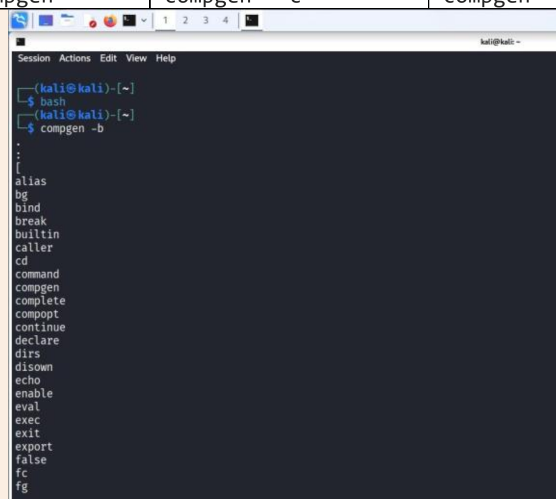
Date:	25 August 2026
Subject:	Linux Operating System Lab

1. Getting Help

Task	Command Name	Syntax	Example	Screenshots
Manual page for a known command	man	man <command>	man mkdir	▼ below
				
Manual page for an unknown command	apropos	apropos	apropos copy	
Know the source file or binary	whereis	whereis <command>	whereis bash	▼ below
				
Know the path of a command	which	which <command>	which ls	▼ below
				
Determine whether command is external or internal	type	type<command>	type ls	▼ below



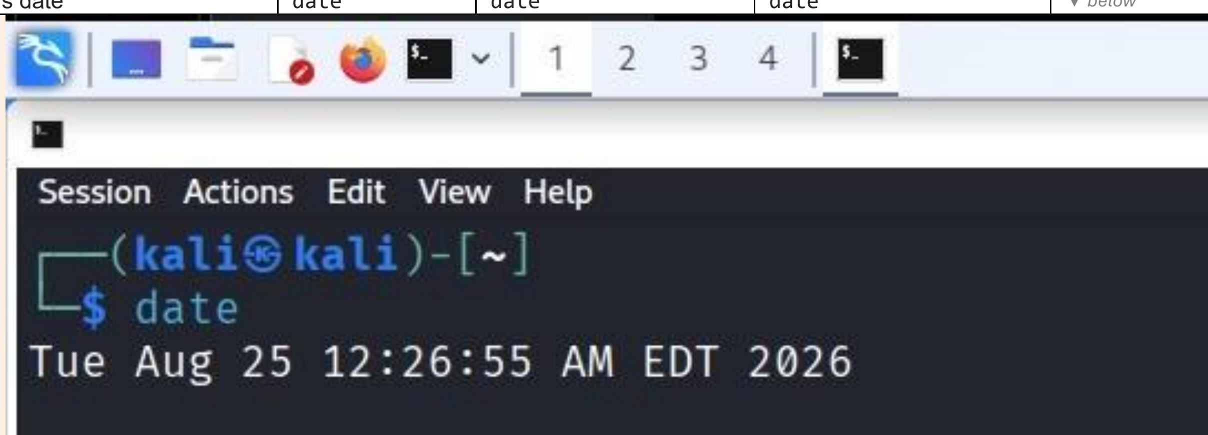
Help for an internal command	help	help <command>	help cd	
List Bash commands	compgen	compgen - c	compgen -b	▼ below



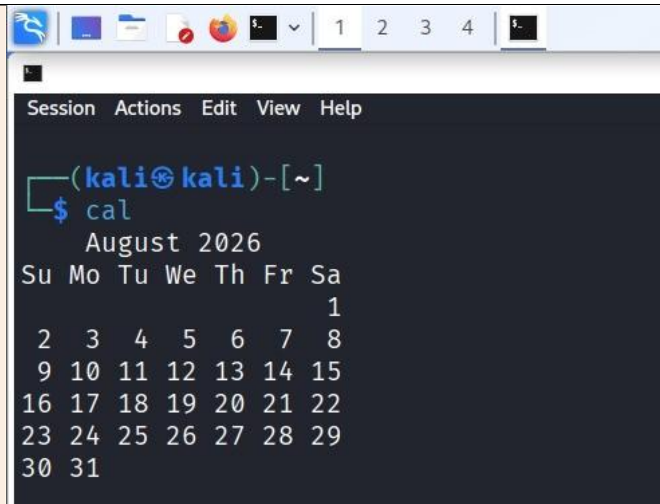
Know command usage	whatis	whatis <command>	whatis ls	

2. Basic Commands

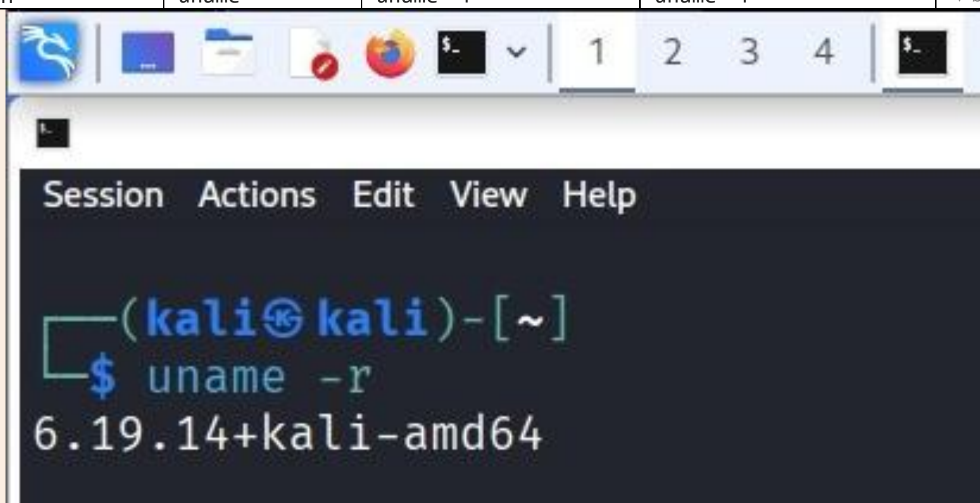
Task	Command Name	Syntax	Example	Screenshots
Today's date	date	date	date	▼ below



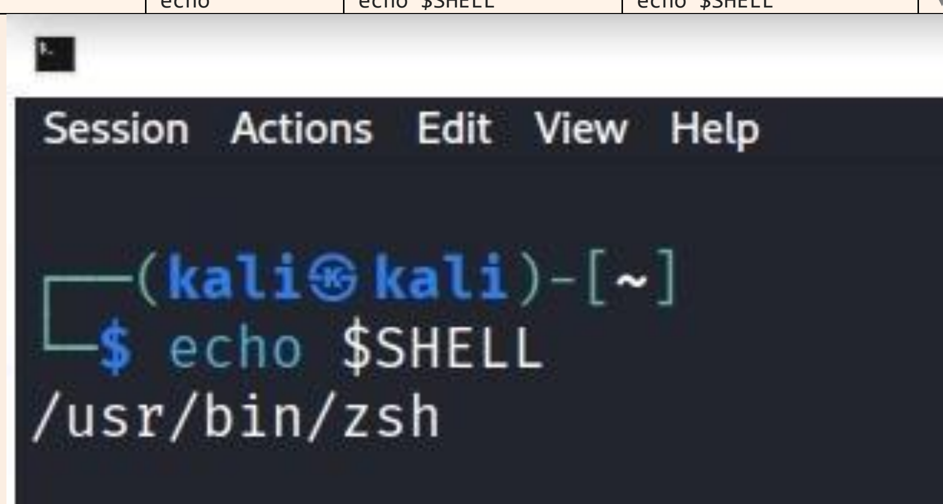
Print calendar	cal	cal [month] [year]	cal	▼ below



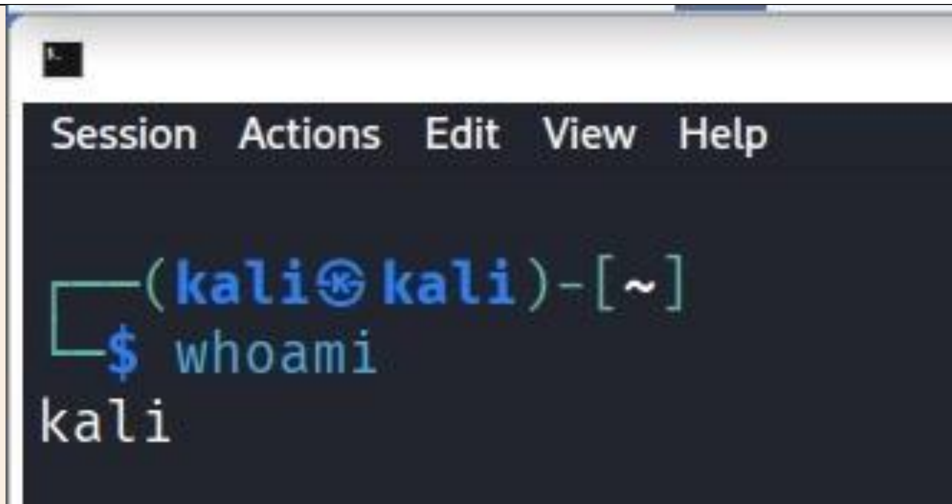
Print kernel version	`uname`	`uname -r`	`uname -r`	▼ *below*



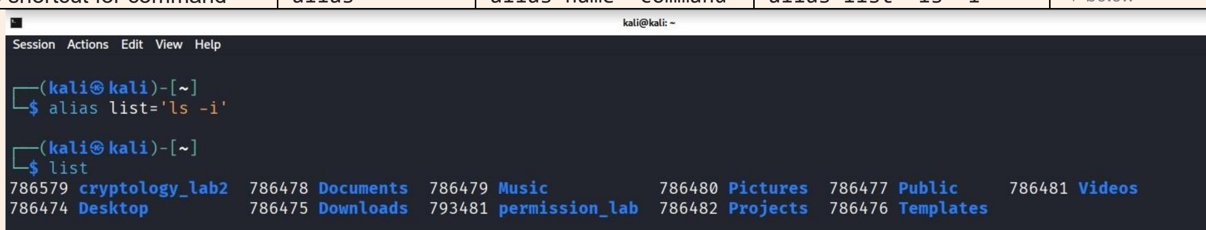
Print default shell	`echo`	`echo $SHELL`	`echo $SHELL`	▼ *below*



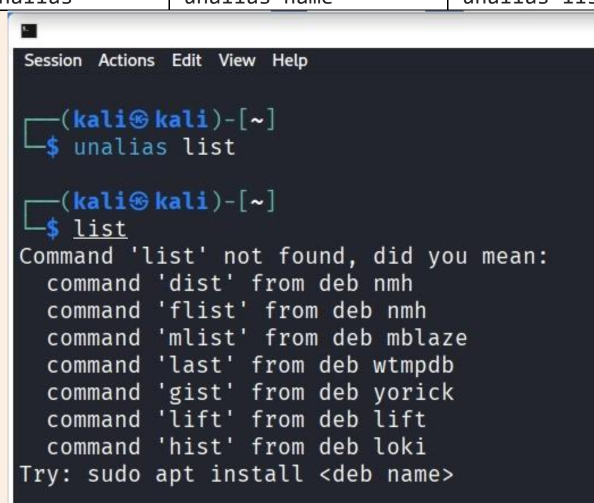
Currently logged-in user	`whoami`	`whoami`	`whoami`	▼ *below*



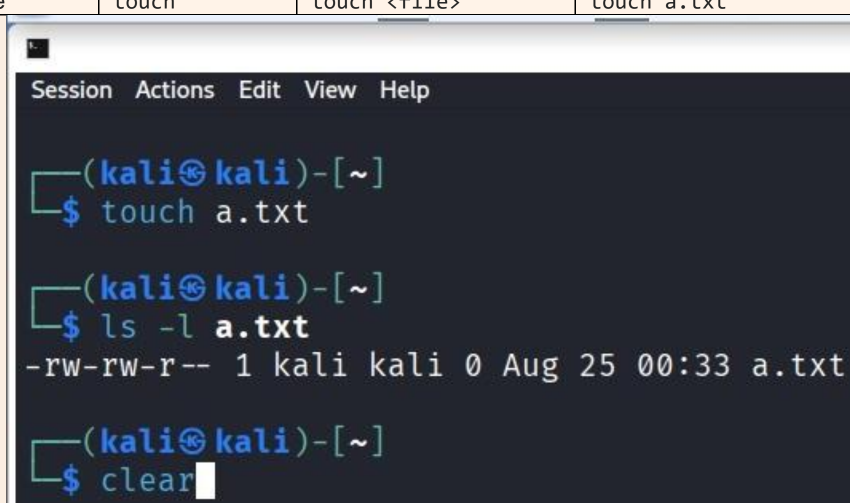
Create shortcut for command	`alias`	`alias name='command'`	`alias list='ls -i'`	▼ below



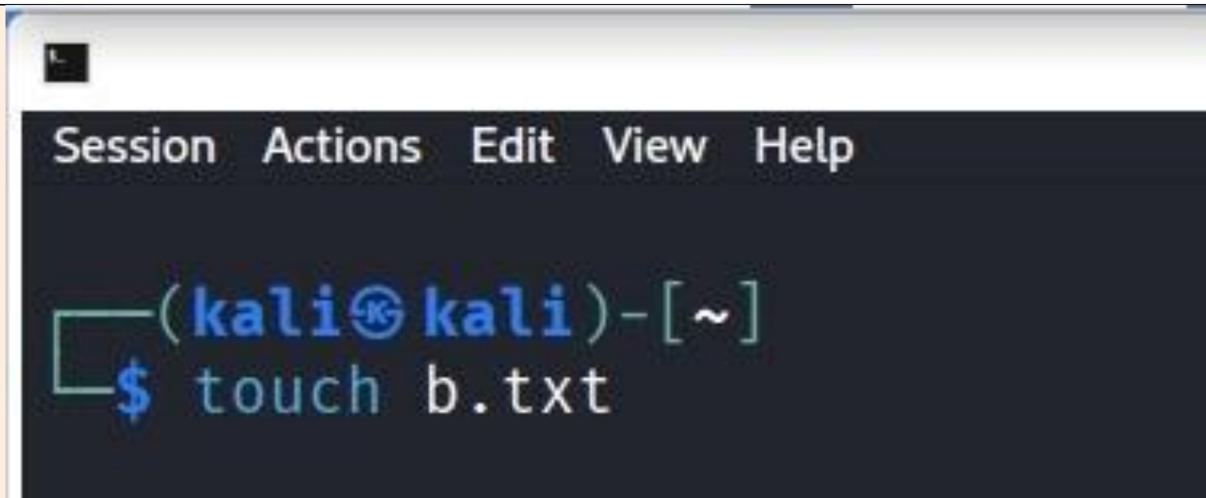
Delete shortcut	`unalias`	`unalias name`	`unalias list`	▼ below



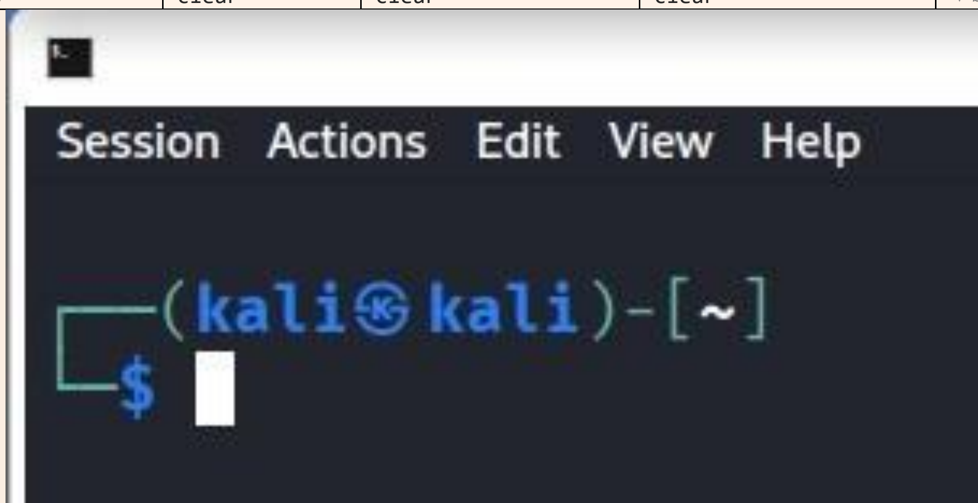
Change timestamp of file	`touch`	`touch <file>`	`touch a.txt`	▼ below



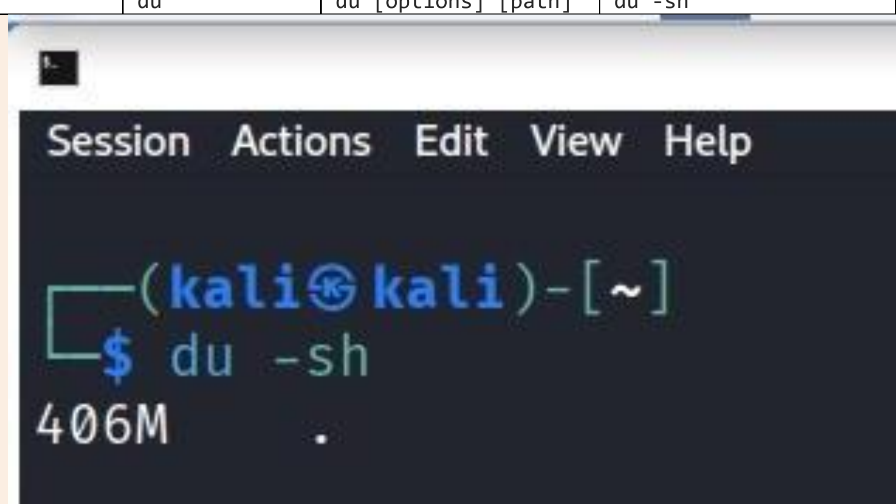
Clear screen	`touch`	`touch file1 file2`	`touch b.txt`	▼ below



Create empty files	`clear`	`clear`	`clear`	▼ *below*



Know disk usage	`du`	`du [options] [path]`	`du -sh`	▼ *below*



Know free space	`df`	`df [options]`	`df -h`	▼ *below*

```
kali@kali: ~
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G     0  1.9G   0% /dev
tmpfs           389M  1.2M  387M   1% /run
/dev/sda1        79G   17G   59G  22% /
tmpfs           1.9G  4.0K  1.9G   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.9G  8.0K  1.9G   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           389M  108K  389M   1% /run/user/1000
```

Linux release information	`cat`	`cat /etc/os-release`	`cat /etc/os-release`	▼ below

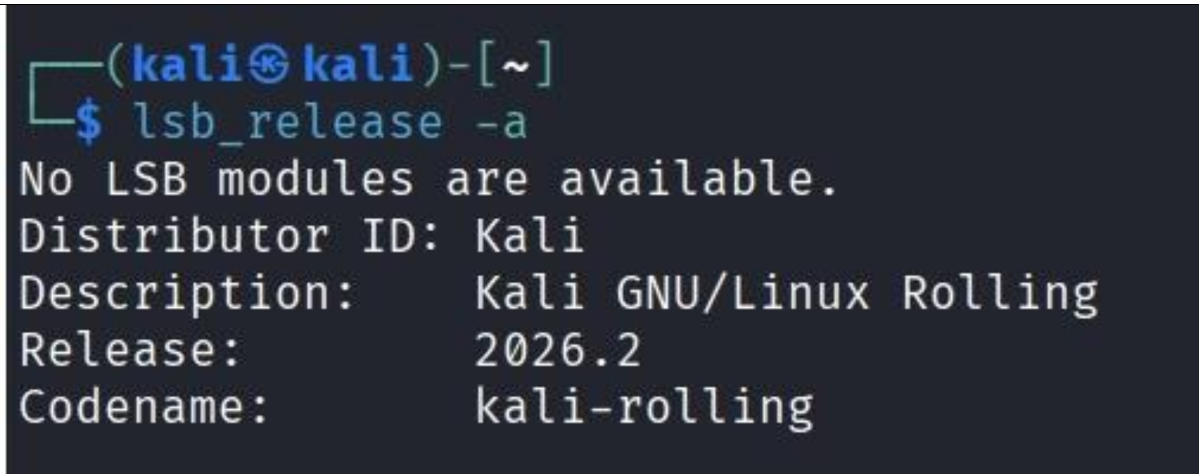
```
┌──(kali㉿kali)-[~]
└─$ cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.2"
VERSION="2026.2"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"
```

Show login history	`last`	`last`	`last`	▼ below

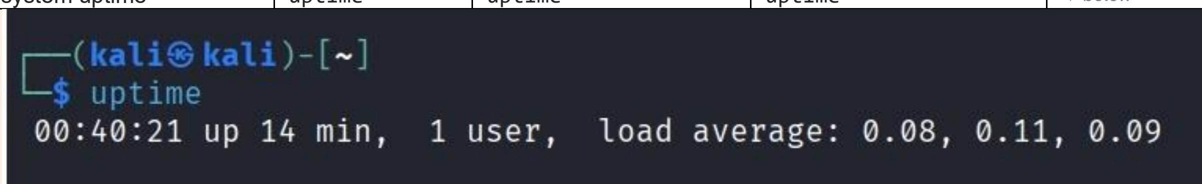
```
┌──(kali㉿kali)-[~]
└─$ last
kali     tty7         :0               Tue Aug 25 00:26 - still logged in
lightdm  tty7         :0               Tue Aug 25 00:26 - 00:26  (00:00)
kali     tty7         :0               Mon Aug 24 13:39 - 14:23  (00:43)
lightdm  tty7         :0               Mon Aug 24 13:39 - 13:39  (00:00)
kali     tty7         :0               Mon Aug 24 09:08 - 09:28  (00:19)
lightdm  tty7         :0               Mon Aug 24 09:08 - 09:08  (00:00)
kali     tty7         :0               Fri Aug 21 04:34 - 06:30 (3+01:55)
lightdm  tty7         :0               Fri Aug 21 04:34 - 04:34  (00:00)
kali     tty7         :0               Fri Aug 21 04:10 - still logged in
lightdm  tty7         :0               Fri Aug 21 04:10 - 04:10  (00:00)
kali     tty7         :0               Wed Aug 19 01:44 - 01:46  (00:02)
lightdm  tty7         :0               Wed Aug 19 01:44 - 01:44  (00:00)
kali     tty7         :0               Tue Aug 18 23:53 - still logged in
lightdm  tty7         :0               Tue Aug 18 23:53 - 23:53  (00:00)
kali     tty7         :0               Tue Aug 18 09:31 - 09:34  (00:03)
lightdm  tty7         :0               Tue Aug 18 09:31 - 09:31  (00:00)
kali     tty7         :0               Tue Aug 18 09:23 - 09:30  (00:07)
lightdm  tty7         :0               Tue Aug 18 09:23 - 09:23  (00:00)
postgres                               Tue Jun 16 10:29 - 10:29  (00:00)

wtmpdb begins Tue Jun 16 10:29:31 2026
```

Check current user privileges	`sudo`	`sudo -l`	`sudo -l`	
Find Linux release	`lsb_release`	`lsb_release -a`	`lsb_release -a`	▼ below

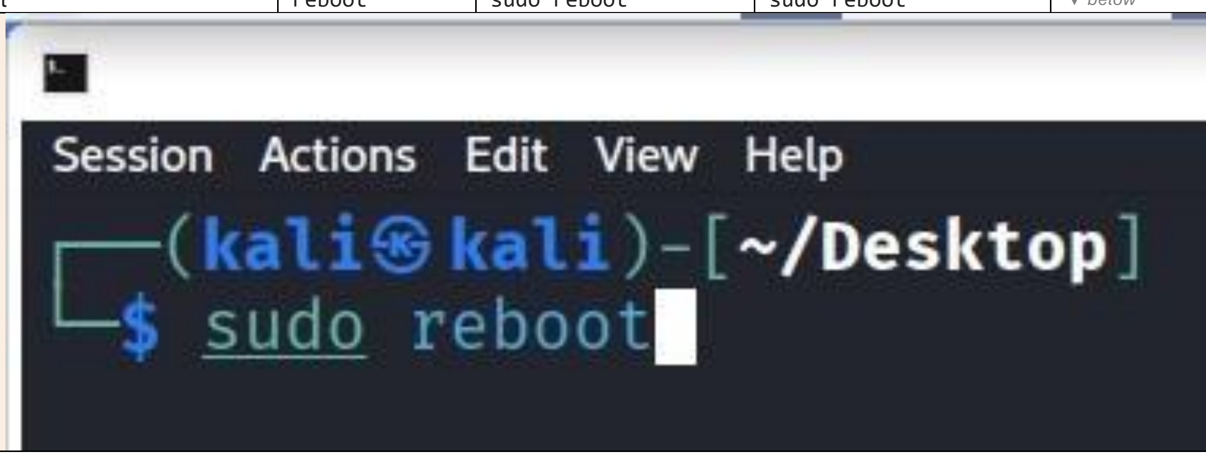


Check system uptime	`uptime`	`uptime`	`uptime`	▼ *below*



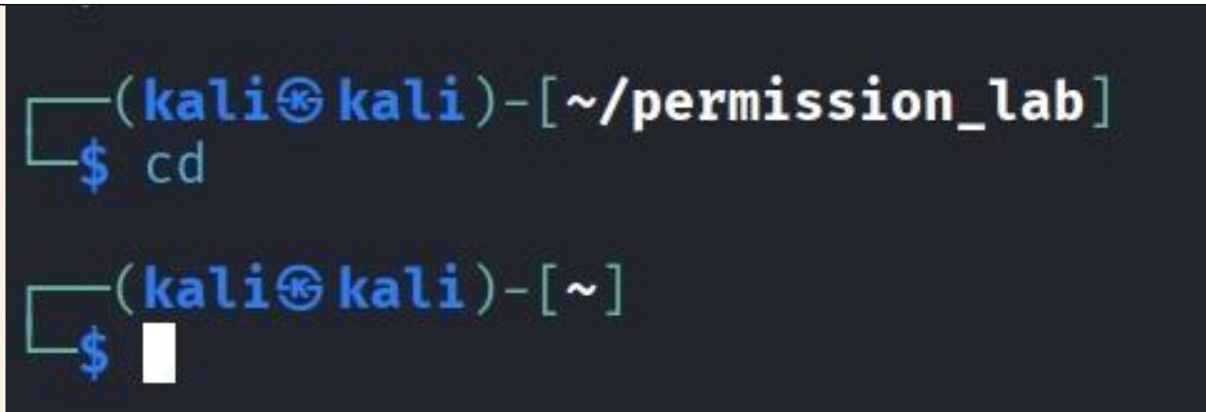
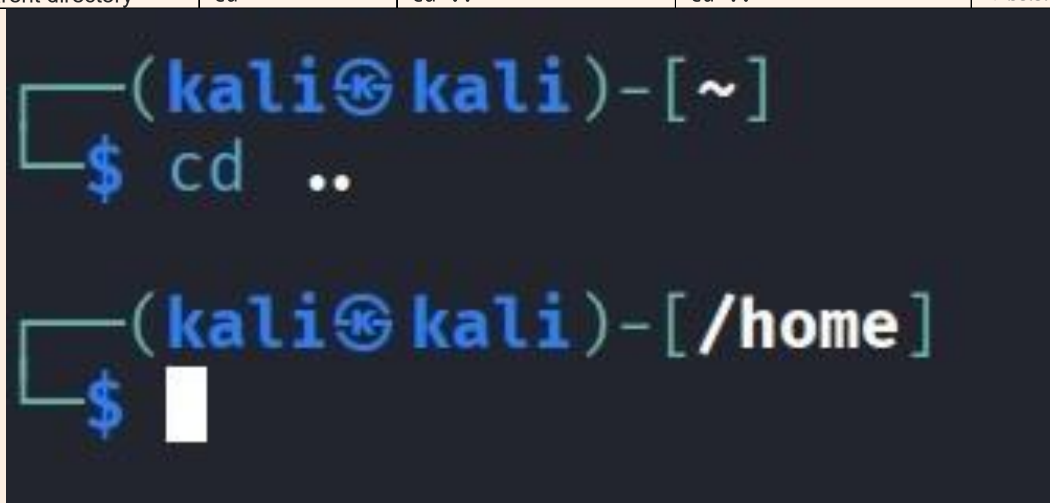
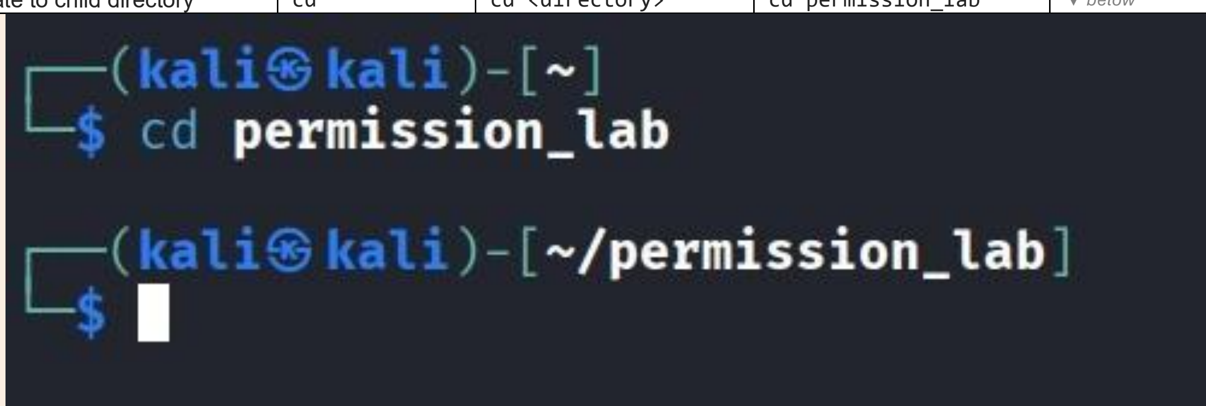
Shutdown	`shutdown`	`sudo shutdown -h now`	`sudo shutdown now`	▼ *below*

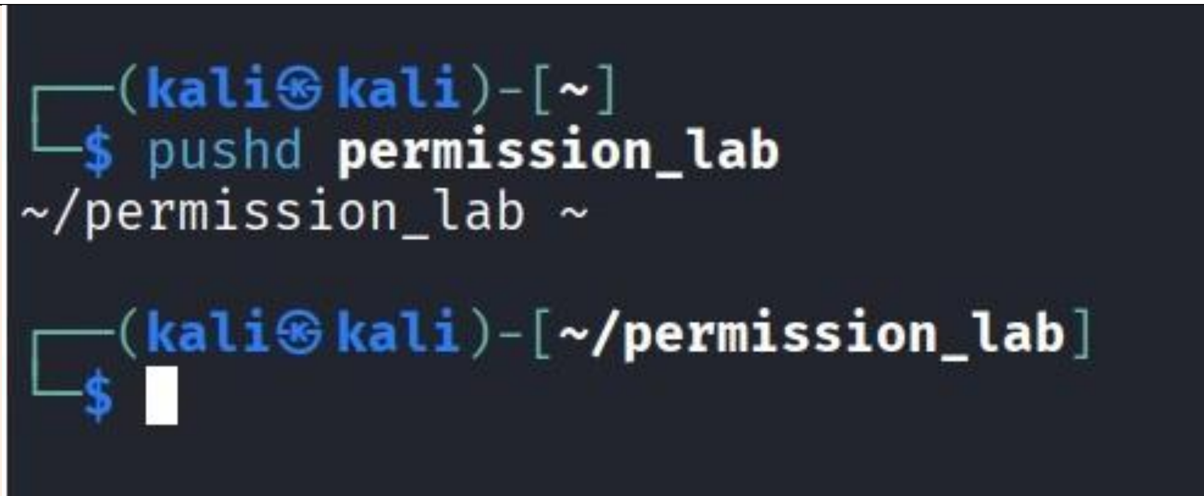
Restart	`reboot`	`sudo reboot`	`sudo reboot`	▼ *below*



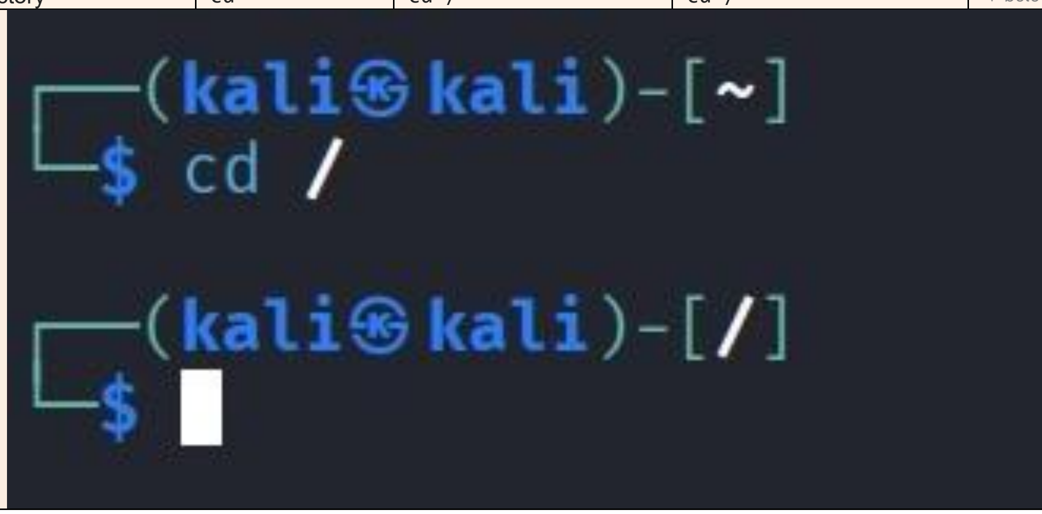
3. Navigation

Task	Command	Syntax	Example	Screenshots
Navigate to home directory	`cd`	`cd ~`	`cd`	▼ *below*

				
Navigate to parent directory	cd	cd ..	cd ..	▼ below
				
Navigate to child directory	cd	cd <directory>	cd permission_lab	▼ below
				
Alternate command to cd	pushd	pushd <directory>	pushd permission_lab	▼ below



Go back to previous directory	cd -	cd -	cd -	
Go to root directory	cd	cd /	cd /	▼ below



4. File System

Task	Syntax	Command / Example
Identify the file system	df -T [path]	df -T .

5. Globbing

Create the practice directory and sample files with the following commands:

mkdir -p ~/glob_practice && cd ~/glob_practice

touch index.html login.html admin.htm report.txt access.log error.log user1.txt user2.log file123 backup1.html backup2.txt

Task	Pattern	Command
a. List all .html files	*.html	ls *.html
b. List all files other than .html files	!(*.html)	ls !(*.html)
c. Names containing at least one digit	*[0-9]*	ls *[0-9]*
d. user followed by a single digit	user[0-9].*	ls user[0-9].*
e. Ending in .txt or .log	+(*.txt|*.log)	ls +(*.txt|*.log)
f. Not ending in .txt or .log	*.{txt,log}	ls *.{txt,log}
g. Beginning with access or error	{access*,error*}	ls {access*,error*}
h. .html files containing a digit	*[0-9]*.html	ls *[0-9]*.html

6. Extended Globbing

Task	Pattern	Command
a1. Select access.log, error.log and auth.log	`@(access.log|error.log|auth.log)`	`ls @(access.log|error.log|auth.log)`
a2. Select all files except notes.txt	`!(@(notes.txt))`	`ls !(@(notes.txt))`
a3. Select auth.log.1 or auth.log.2	`auth.log.@(1|2)`	`ls auth.log.@(1|2)`
b. malware .log names containing a digit	`malware*[0-9]*.log`	`ls malware*[0-9]*.log`
c1. .log files containing failed_login or successful_login	`@(failed_login|successful_login).log`	`ls @(failed_login|successful_login).log`
c2. All .log files except firewall.log	`!(@(firewall.log))`	`ls !(@(firewall.log))`

7. Filters

Use the supplied Lab2_log_files directory for these exercises. Verified archive contents: login.log (50 lines), access.log (80 lines), security.log (120 lines), and incident.log (500 lines).

Item	Command
a. First 10 entries of login.log	`head -n 10 login.log`
b. First 5 entries of access.log	`head -n 5 access.log`
c. First 15 security events	`head -n 15 security.log`
d. First 20 incident events	`head -n 20 incident.log`
e. Only first entry in login.log	`head -n 1 login.log`
f. First 25 web requests	`head -n 25 access.log`
g. Last 10 login entries	`tail -n 10 login.log`
h. Last 5 access entries	`tail -n 5 access.log`
i. Last 20 security events	`tail -n 20 security.log`
j. Last 25 incident events	`tail -n 25 incident.log`
k. Only last security entry	`tail -n 1 security.log`
l. Last 50 incident events	`tail -n 50 incident.log`
m. login.log from line 41	`tail -n +41 login.log`
n. security.log from line 101	`tail -n +101 security.log`
o. Total lines in login.log	`wc -l login.log`
p. Total lines in access.log	`wc -l access.log`
q. Total security events	`wc -l security.log`
r. Total incident events	`wc -l incident.log`
s. Number of words in login.log	`wc -w login.log`
t. Number of bytes in security.log	`wc -c security.log`
u. Line, word, byte counts of access.log	`wc -l -w -c access.log`
v. Line counts of all four log files	`wc -l *.log`

Item	Command
w. First 20, then last 5	`head -n 20 login.log	tail -n 5`
x. Last 20, then first 5	`tail -n 20 login.log	head -n 5`
y. First 30 access, then last 10	`head -n 30 access.log	tail -n 10`
z. Last 30 access, then first 10	`tail -n 30 access.log	head -n 10`
aa. First 50 security, then last 10	`head -n 50 security.log	tail -n 10`
bb. Last 50 security, then first 10	`tail -n 50 security.log	head -n 10`
cc. login lines 11–20	`sed -n "11,20p" login.log`
dd. login lines 21–30	`sed -n "21,30p" login.log`
ee. access lines 11–20	`sed -n "11,20p" access.log`
ff. access lines 31–40	`sed -n "31,40p" access.log`
gg. security lines 21–30	`sed -n "21,30p" security.log`
hh. security lines 51–60	`sed -n "51,60p" security.log`
ii. incident lines 101–110	`sed -n "101,110p" incident.log`
jj. incident lines 201–220	`sed -n "201,220p" incident.log`
kk. incident lines 491–500	`sed -n "491,500p" incident.log`
ll. Count first 20 login entries	`head -n 20 login.log	wc -l`
mm. Count last 15 access entries	`tail -n 15 access.log	wc -l`

nn. Count words in first 50 security entries	`head -n 50 security.log | wc -w`
oo. Count lines in last 25 security entries	`tail -n 25 security.log | wc -l`
pp. Count extracted login lines 21–30	`sed -n "21,30p" login.log | wc -l`
qq. Count extracted access lines 51–60	`sed -n "51,60p" access.log | wc -l`
rr. Count incident events 101–150	`sed -n "101,150p" incident.log | wc -l`
ss. First 100 incident, last 20, count	`head -n 100 incident.log | tail -n 20 | wc -l`